

Edge Detection in Images Using MATLAB

Author: Safa Bazrafshan

Email: safa.bazrafshan@gmail.com

GitHub: github.com/safa-bazrafshan

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1. Project Overview

Edge detection is a fundamental technique in image processing and computer vision. This project demonstrates how to apply popular edge detection algorithms using MATLAB, including Sobel, Prewitt, and Canny, to detect object boundaries in a grayscale image.

2. Tools Used

MATLAB R2023a

Image Processing Toolbox

3. Steps Performed

a) Load and Display the Original Image

The built-in `peppers.png` image was used as the input. It contains colorful objects with well-defined edges.

b) Convert to Grayscale

Using `rgb2gray()`, the color image was converted to grayscale to simplify processing.

c) Apply Edge Detection

We used MATLAB's `edge()` function with different methods:

Sobel – emphasizes horizontal and vertical edges

Prewitt – similar to Sobel with slightly different masks

Canny – multi-stage algorithm, better noise suppression

4. Results

Below are visual results of the filters applied:

Original Image

Grayscale Image

Edge Detected Images using Sobel, Prewitt, and Canny

Original Image



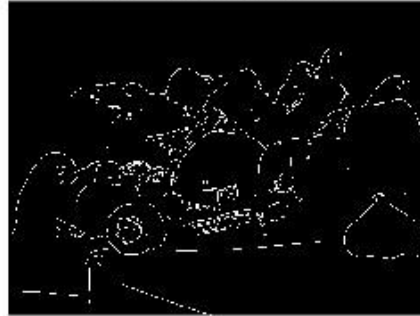
Grayscale Image



Grayscale Image



Sobel Edge Detection



Prewitt Edge Detection



Canny Edge Detection



5. Conclusion

Edge detection provides a strong base for object detection, segmentation, and image analysis. Among the methods, Canny generally gives the most accurate edges with reduced noise.

6. Future Improvements

Try edge detection on video frames

Combine with object detection algorithms

Experiment with noise-affected images

7. References

MATLAB Documentation – <https://www.mathworks.com/help/images/edge.html>

Digital Image Processing by Gonzalez and Woods